A Multi-Level Data-Driven Analysis of Road Traffic Accidents in India

Geographical, Temporal, Seasonal, Environmental and Severity Assessment | B.Tech ECE, LPU | MoRTH 2018-2024

PROBLEM + DATA (254 state-years, 38 entities, 236 weather matched)

2024: 487,707 accidents | 177,175 fatalities | 471,441 injured (India-level).

India +3.7% acc / +12.4% fat; ratio 33.50>36.33; 2020 dip+recovery.

Friedman X2=25.29 p=3.0e-4 W=0.72 (no ranking).

States acc 17+/15-/6~, fat 26+/10-/2~.

KEY FINDINGS

Weather: rain r=-0.234/-0.376; Tmax +0.452/+0.498; Tmin +0.366/+0.340 (assoc. only).

NCRB 2024: May 8.84%, Jul 7.46%, X2=1276 (separate vintage).

Overspeeding 70.7%/70.3% (reported). Hit from Back frequent; Hit-and-Run ratio ~0.50.

Two-wheelers 46.2%, pedestrians 20.6% (shares, not risk). Valid licence 68.2%.

ML + CLUSTERING + APP

ML: train 2019-22 N=141, test 2023-24 N=72. Naive RMSE 1038.6/286.3 BEATS all trained (honest nu

K-means K=2 silhouette 0.356, sizes 17/6, descriptive only.

Limits: no microdata/coords, no vehicle stock/AADT, 2023-24 state injuries N/A.

Live: https://morth-accident-analytics.onrender.com

Primary: MoRTH 2018-2024 | Supporting: MoRTH 2024, NCRB ADSI 2024, IMD | Conclusion: reproducible framework; associations not causes; counts not risk.

Live Analytics Platform